

Twitter Therapy: Utilizing Natural Language Processing to Detect Depression in Tweets

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Abstract

With the rise in popularity of social media networks in recent years, the world has seen an unprecedented amount of online user-generated data. With such an increase in volume of information on our hands, we ask ourselves if it is possible to use Tweets to detect public health issues, namely depression, in online users. We identify textual patterns indicative of depressive behavior in tweets by preprocessing and labeling data, performing feature extraction sensitive to context, and employing three pre-trained BERT models: BERT classifier on dataset without modification, BERT classifier on dataset with the term 'depression' removed, and BERT Tokenizer with Random Forest Classifier. Based on our results, we found that the BERT classifier trained on our dataset without modification achieved the highest accuracy of 84.95% of our three models, as well as the highest precision, recall, and F1-scores. With this study, we hope to demonstrate the role NLP can play in depression's early detection, bringing awareness to how user-generated data can contribute to the realm of healthcare.

1 Introduction

Depression, being one of the most common mood disorders worldwide, affects around 300 million people, with 35% of diagnosed adults receiving no treatment despite there existing numerous treatment plans for the condition. With such a large percentage of those diagnosed not seeking treatment, how do we prevent an undiagnosed individual from doing the same? Early detection of depression is thus crucial for effective intervention and treatment, and our study aims to utilize our abundance of online user-generated data to develop a machine learning model that can successfully detect depression in users through their Tweet content.

With online users often turning to social media as an emotional outlet, our project intends to test

the theory that an individual's social media content can be indicative of their mental health state. We utilize a dataset of Tweets labeled 'depressed' or 'non-depressed' to indicate if the user has been diagnosed with depression, and perform machine learning techniques to train our models and subsequently test how accurately our models can detect if a user is depressed from a Tweet. We aim to unfold the way a user's mental state can present itself in language, particularly in social media spaces that are easily accessible as public data.

We use a combination of data preprocessing, context-sensitive feature extraction, and the use of pre-trained BERT models for sophisticated natural language processing. As BERT is a highly advanced, capable language model that can detect the intricacies in language of our dataset, we use three different pre-trained models: BERT classifier on our dataset with no modifications, BERT classifier on our dataset after removing the word 'depression' as it is highly overrepresented, and BERT tokenizer with Random Forest Classifier.

The elimination of the term 'depression' from our dataset for our second pre-trained BERT model is to train the model to focus on contextual cues and broader language patterns rather than relying on specific keywords, as would be the case in real life scenarios. In addition, we address potential biases and ethical considerations by avoiding demographic labeling in favor of clustering techniques that provide non-intrusive user categorization.

For evaluation, we test model performance across different subsets of data using k-fold cross-validation. We use the performance metrics of accuracy, precision, recall, and F1 score, with a focus on precision and recall. We also generate a confusion matrix for a visual representation of our models' predictive accuracy.

Our research, by incorporating the potential of social media data for mental health analysis, holds significant implications for public health. It of-

fers an efficient approach to identifying signs of depression, inciting early intervention and treatment strategies, contributing to the general efforts in public health care and awareness.

2 Related Works

For our research, three studies stand out as particularly relevant, offering insights closely aligned with our goals:

Sentiment Analysis: It's Complicated! discusses the tendency of dataset constructors to discard “noisy” or “controversial” data where there is significant disagreement on the proper label. In the context of datasets containing Tweets, such controversial data can comprise over 30% of the original data. The paper argues that the discarding of this data is problematic, as models cannot detect in real-time whether or not a sample will fall into this controversial category and thus, should be trained on the data regardless under the label of a “complicated” class of sentiment. Our study is informed by this study as it offers insights into how our data should be labeled, and how the labeling of our data can subsequently impact the performance of our models.

Towards Preemptive Detection of Depression and Anxiety in Twitter's objective aligns closely with ours, as it aims to detect depression and anxiety in tweets from non-diagnosed users. Its findings can help us build upon our models as it emphasizes the challenges of unbalanced training sets and the complicated application of language cues. We can make use of these findings for optimizing our models' performances in detecting depression indicators.

Sentiment Analysis in Social Media Data For Depression Detection Using Artificial Intelligence: A Review aims to provide a more multifaceted understanding of social media data by using multi-class sentiment analysis. It found that multi-class classification yielded a higher precision value during sentiment analysis as opposed to binary or ternary classification. By using machine learning and deep learning techniques, this study demonstrates NLP's role in analyzing complex emotional landscapes as indicated across tweets, and provides direction for our project in achieving higher precision by also accurately identifying complex emotional expressions.

These studies that we analyzed through our literature review provide us with a foundation which

we can build upon to optimize our models. They offer an invaluable introduction to the field of NLP as an early detector of mental health issues, presenting us with different methodologies and techniques. Moving forward with our own research, we will draw inspiration from the research conducted before us to showcase the impact NLP can have on intervention in public health.

3 Data

3.1 Dataset

This study utilized 20,000 real tweets sourced from Kaggle, specifically designed for depression detection through analysis. The dataset consists of a collection of tweets drawn from a variety of user accounts and is labeled either with a 1, indicating depression in the user who posted it, or 0 otherwise. Tweets labeled with the depression tag were drawn from accounts who had a tweet proclaiming that they had been medically diagnosed with depression.

Of the 20,000 tweets, 14,000 were selected for the development set, 20% of which was used as validation data. Of these tweets, 50% of the tweets were labeled with the depression tag and the other 50% was not. Our test set consisted of 6,000 tweets, which also had a fifty-fifty split of depressed and non-depressed tags.

3.2 Limitations

Our dataset posed some challenges, one of which was that the multitude of posts from each individual user was labeled with the same depression indicator. This made it difficult to track progression or variation in the user's mental state over time, and limited the ability to observe shifts in sentiment across posts. Thus, we were unable to use techniques like recurrent neural networks, which typically are effective in capturing changes over time.

3.3 Processing

We began the task of data processing by cleaning up tweet data. This included removing retweet headings and URLs from the text, then removing stop words and special characters, and finally converting all words to lowercase.

Then, we conducted word occurrence ratio analysis on the frequency of key words in the labeled and unlabeled data. Significant disparities were found among the occurrence ratios for different

Word: 'anxiety', Ratio: 20.20
 Word: 'depression', Ratio: 110.75
 Word: 'symptom', Ratio: 29.00
 Word: 'illness', Ratio: 59.00
 Word: 'mental', Ratio: 27.60
 Word: 'addiction', Ratio: 82.00
 Word: 'suicide', Ratio: 21.67
 Word: 'disorder', Ratio: 21.33
 Word: 'benefit', Ratio: 29.00
 Word: 'treatment', Ratio: 175.00
 Word: 'autism', Ratio: 48.00
 Word: 'overcome', Ratio: 109.50
 Word: 'awareness', Ratio: 27.00
 Word: 'motivation', Ratio: 20.50
 Word: 'negative', Ratio: 30.50
 Word: 'cow', Ratio: 26.00
 Word: 'meat', Ratio: 24.00
 Word: 'relief', Ratio: 21.00
 Word: 'remedy', Ratio: 37.00
 Word: 'article', Ratio: 24.50

Figure 1: Word Occurrence Ratios

words as seen in Figure 1, providing insight on their potential relevance within the scope of depression detection. The words with abnormally high ratios were typically associated with mental health problems, such as ‘depression’ and ‘addiction’, as well as words related to the management of these issues, e.g. ‘treatment’ and ‘overcome’.

To address the over-representation of the word "depression" in depression-labeled tweets, we implemented a strategy of randomly deleting the word "depression" from these posts - whenever the word "depression" was encountered in a tweet, there was a 99% chance that it would be deleted.

The goal of this deletion strategy was to reduce the model’s dependency on explicit mentions of ‘depression’, allowing it to focus more on subtle, context-based indicators to classify each tweet. Therefore, the model is better suited to real-world applicability by enabling it to identify a broader range of indicators without having to rely on explicit mentions of depression.

Dep. Variable:	label	No. Observations:	20000
Model:	Logit	Df Residuals:	19998
Method:	MLE	Df Model:	1
Date:	Thu, 14 Dec 2023	Pseudo R-squ.:	0.04000
Time:	05:05:00	Log-Likelihood:	-13308.
converged:	True	LL-Null:	-13863.
Covariance Type:	nonrobust	LLR p-value:	3.792e-243
=====			
	coef	std err	z
const	-0.0865	0.014	-5.981
treatment	4.6191	0.335	13.771
=====			
			P> z
			[0.025
			0.975]
const	-0.0865	0.014	-5.981
treatment	4.6191	0.335	13.771
=====			

Figure 2: ATE of term "depression" before random removal

Dep. Variable:	label	No. Observations:	20000
Model:	Logit	Df Residuals:	19998
Method:	MLE	Df Model:	1
Date:	Thu, 14 Dec 2023	Pseudo R-squ.:	0.04000
Time:	05:05:00	Log-Likelihood:	-13308.
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Covariance Type:	nonrobust	LLR p-value:	3.792e-243
=====			
	coef	std err	z
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treatment	4.6191	0.335	13.771
=====			
			P> z
			[0.025
			0.975]
const	-0.0865	0.014	-5.981
treatment	4.6191	0.335	13.771
=====			

Figure 3: ATE of term "depression" after random removal

Through analyzing the ATE (average treatment effect) using basic logistical regression, the dataset before and after the random removal of the term "depression" from tweets labeled as indicative of depression show a notable change in the model’s dependency on this specific term, . Initially, the ‘treatment’ coefficient was 4.6191, signifying a heavy reliance on the presence of "depression" for classification, with the model explaining about 4% of the variance in the outcome. After the term’s removal, this coefficient decreased to 1.8147, and the Pseudo R-squared value dropped to 0.001329 from 0.04, indicating a lesser degree of variance explained by the model. This reduction in both the coefficient and the explanatory power of the model suggests that with the key term "depression" omitted, the basic logistical model’s predictive reliance on this single word is reduced, possibly leading to a more generalized approach to classification that is less susceptible to overfitting and better at identifying depression based on a broader range of textual indicators.

4 Methodology

Our study employed two methodologies to analyze Twitter data for indications of depression. The first approach utilized BERT (Bidirectional Encoder Representations from Transformers) as a tokenizer, transforming tweets into feature vectors and then providing the vectors to a Random Forest Model. The second used BERT as a standalone classifier, taking advantage of its ability to interpret the context of words in a sentence not just individually, but in the entire context of a sentence or paragraph to understand each word’s meaning and usage.

4.1 BERT Tokenizer with Random Forest Model

The initial approach we took harnessed BERT as a tokenizer in conjunction with a Random Forest classifier. We used BERT’s built-in tokenizer to

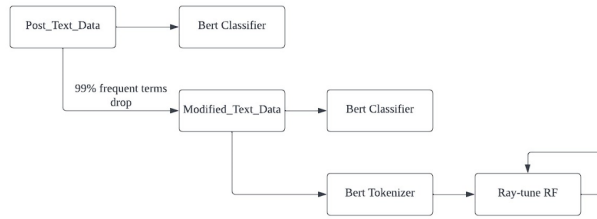


Figure 4: Methodology

transform tweets into vector representations imbued with contextual features. These vectors were then provided as input to a Random Forest classifier, which was chosen for its effectiveness in managing high-dimensional data and its robustness against overfitting.

This hybrid model was trained on the modified dataset, on which the overrepresentation of the word "depression" was mitigated through the random deletion strategy as discussed in the data section. To fine-tune the hyperparameters of the Random Forest classifier, we employed Ray Tune to optimize model performance through bayesian search with different hyperparameter configurations. The performance of this hybrid model was thoroughly assessed over various splits of data, confirming its effectiveness across a range of samples.

4.2 Standalone BERT Classifier

After conducting error analysis on our first approach, we decided to try to improve our results by using a stand alone BERT classifier. We started by preprocessing tweets and their corresponding labels to meet the input criteria of the BERT model. This involved tokenizing the tweets, adjusting sequence lengths, and formatting labels to suit binary classification tasks. Subsequently, the resulting data was fed into the BERT model and then fine-tuned. We then proceeded to validate its performance across different data subsets.

As part of this implementation, we also decided to test the effectiveness of the random deletion strategy. We first trained the BERT model on the original data set without the emission of the word "depression" as a baseline comparison. Then, after the 99% random deletion, we retrained the model on the modified data set to evaluate its ability to identify depression indicators without dependency on a keyword.

5 Results

5.1 Initial Results

Test Accuracy: 0.7613333333333333

	precision	recall	f1-score	support
0	0.75	0.77	0.76	2983
1	0.77	0.75	0.76	3017
accuracy			0.76	6000
macro avg	0.76	0.76	0.76	6000
weighted avg	0.76	0.76	0.76	6000

```

[[2305  678]
 [ 754 2263]]

```

Figure 5: Bert Tokenizer + Random Forest Model Results

The results of the BERT tokenizer with Random Forest approach can be seen in Figure 5. Overall, this implementation had an accuracy ranging from 70% - 76% with different model settings, with a peak accuracy of 76.13%. The model had a relatively balanced precision and recall score, with a precision of 75% and 77% recall on tweets with the non-depressed label, and a 77% precision and 75% recall on tweets with the depressed label. The f-scores for both labels were a balanced 0.76. Furthermore, the confusion matrix reveals that out of 6,000 tweets, there were 754 false negatives and 678 false positives.

While the 76% accuracy of our system is certainly not exceptional, these results are indicative of the potential of our system, as it far outperforms the baseline system of blind guess, which is expected to yield an accuracy of 50%.

5.2 BERT Classifier Results

Compared to the BERT tokenizer and Random Forest Model, the results of the standalone BERT classifier model reveal a significant improvement in performance metrics.

Test Accuracy: 0.8495

Classification Report

	precision	recall	f1-score	support
Class 0	0.86	0.84	0.85	2983
Class 1	0.84	0.86	0.85	3017
accuracy			0.85	6000
macro avg	0.85	0.85	0.85	6000
weighted avg	0.85	0.85	0.85	6000

Confusion Matrix

```

[[2504  479]
 [ 424 2593]]

```

Figure 6: BERT Classifier Results Without Random Term Removal

As seen in figure 6, without the omission of the term 'depression', the model achieved an accuracy of 84.95%, with a balanced precision and recall score ranging from 84% to 86% for both tweets labeled with the depression tag and tweets without it. The f-score was 0.85 for both classes, and the confusion matrix demonstrates a drop to only 424 false negatives and 479 positives.

Test Accuracy: 0.8456666666666667				
Classification Report				
	precision	recall	f1-score	support
Class 0	0.85	0.84	0.84	2983
Class 1	0.84	0.85	0.85	3017
accuracy			0.85	6000
macro avg	0.85	0.85	0.85	6000
weighted avg	0.85	0.85	0.85	6000
Confusion Matrix				
[[2497 486]				
[440 2577]]				

Figure 7: BERT Classifier Results With Random Term Removal

However, perhaps even more promising are the results of BERT classifier with 99% 'depression' term removal. The overall accuracy was 84.57%, with a precision and recall range of 84% to 85% for both tags, and an f-score of 0.84 for tweets tagged non-depressed, and 0.85 for tweets tagged depressed. In total, there were 440 false negatives, and 486 false positives.

These numbers provide a multitude of insights, the first being that even without the inclusion of the key term 'depression', the system is still able to effectively classify tweets. There was only a 0.38% drop in accuracy, and the precision, recall, and f-scores stayed relatively consistent proving that the model does not rely on explicit mentions of depression.

Furthermore, this alternative algorithm beat out our initial tokenizer and random forest approach by a narrow margin.

6 Conclusion

Our project utilized post texts data to create classification models to detect depression for early interventions and potential treatment in mental health. By employing deep learning transformers; specifically, we developed a system that can analyze tweet content for indications of depressive behaviors with a pre-trained BERT model and Random Forest Classifier.

The BERT pre-trained transformer model is used in both direct and indirect references to depression.

When it serves as a standalone classifier, it demonstrates a high level of accuracy, precision, recall, and F1-score, suggesting that it has considerable performance with the available data and is capable of capturing the subtleties of language indicative of depression. Meanwhile, it also shows a sign of overfitting. In comparison, the BERT tokenizer with a Random Forest Classifier, while effective, did not match the performance of the standalone BERT model. The performance gap highlights the advantages of deep learning in extracting and processing complex patterns within text data.

One of our most significant findings is the BERT model's robust performance despite the removal of the explicit term "depression", which has a strong correlation with the label. This indicates the model's capacity to learn from a broad range of linguistic cues, rather than overfitting to specific keywords, which is essential for real-world applications where depressive language may be implicit.

However, the observed difference between training and test accuracy for the BERT classifier suggests that overfitting may be a concern, and future iterations of the model should incorporate strategies to mitigate this risk.

7 Future Work

For enhancement of our research, there are several modifications we can conduct to further optimize our model's capabilities. Namely, our dataset can be expanded to include more Tweets that represent the diversity of language that can be related to depression. This will improve our model's generalisability, allowing it to be applied to more scenarios and larger volumes of different samples. Expanding our horizons, we can take our research a step further by incorporating text data found on other social media sites such as Facebook, Tumblr, etc. that will allow our model to be highly applicable to many different cases of user-generated text content.

In addition, we can incorporate sentiment analysis into our research, particularly multi-class sentiment analysis which can highlight different emotional aspects of depression and may lead to higher precision as proven by other multi-class sentiment research projects. This will expand our model's ability to identify a user's emotional state and better represent the plethora of sentiments expressed on social media. This modification can help us segue into identifying not only depression, but other mood disorders as well, building upon our model's

contribution to intervention in public health issues as a powerful tool for potential diagnosis.

Furthermore, our current dataset does not take into account a Tweet’s context, particularly if it is related to any other Tweets made by the same user and when it was published. By looking at a user’s history of Tweets, we can gain insight into their emotional state overtime and watch the progression of a mental health condition, potentially drawing a timeline of the disorder which can help us pinpoint more accurate early detection benchmarks. This can aid in scientific research on mental health conditions as well, giving professionals an opportunity to view the evolution of a condition through a vast collection of public data.

Although our research currently attempts to avoid categorization of users based on demographics (as due to ethical concerns), in the future it may be useful to take demographics into consideration so we can see how a user’s identity and background affects the presentation of a mental health disorder online. This allows us to provide more accurate early detection with context provided to the user, but ethical concerns still remain as it is likely for this modification to introduce complications and biases to our research.

Lastly, our project would greatly benefit from the collaboration of mental health professionals who can ensure that our results are aligned with the clinical understanding of depression. Their knowledge is crucial to the effectiveness and accuracy of our work, as having a professional onboard will help us verify the validity of our process. There are several avenues our research can take in the future to become an even more powerful tool to the medical community, seeing the efficiency it currently provides in identifying potential danger signs of depression.

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References

- [1] Kian Kenyon-Dean, Eisha Ahmed, Scott Fujimoto, Jeremy Georges-Filteau, Christopher Glasz, Barleen Kaur, Auguste Lalande, Shruti Bhandari, Robert

Belfer, Nirmal Kanagasabai, Roman Sarrazingen-dron, Rohit Verma, and Derek Ruths. 2018. Sentiment Analysis: It’s Complicated!. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, pages 1886–1895, New Orleans, Louisiana. Association for Computational Linguistics.

- [2] H. Andrew Schwartz, Johannes Eichstaedt, Margaret L. Kern, Gregory Park, Maarten Sap, David Stillwell, Michal Kosinski, and Lyle Ungar. 2014. Towards Assessing Changes in Degree of Depression through Facebook. In *Proceedings of the Workshop on Computational Linguistics and Clinical Psychology: From Linguistic Signal to Clinical Reality*, pages 118–125, Baltimore, Maryland, USA. Association for Computational Linguistics.
- [3] David Owen, Jose Camacho-Collados, Luis Espinosa-Anke. 2020. Towards Preemptive Detection of Depression and Anxiety in Twitter. In *Proceedings of the 5th Social Media Mining for Health Applications (SMM4H) Workshop & Shared Task*, pages 82-89, Barcelona, Spain (Online).
- [4] Babu, N.V., Kanaga, E.G.M. Sentiment Analysis in Social Media Data for Depression Detection Using Artificial Intelligence: A Review. *SN COMPUT. SCI.* 3, 74 (2022). <https://doi.org/10.1007/s42979-021-00958-1>
- [5] Ted Pedersen. 2015. Screening Twitter Users for Depression and PTSD with Lexical Decision Lists. In *Proceedings of the 2nd Workshop on Computational Linguistics and Clinical Psychology: From Linguistic Signal to Clinical Reality*, pages 46–53, Denver, Colorado. Association for Computational Linguistics.